

Demo: Loading Macros via include shortcode

This page demonstrates the [include shortcode](#) strategy for loading macros into a Quarto document.

The `{< include macros.qmd >}` shortcode at the top of this file embeds the entire contents of `macros.qmd` inline at that point. Since `macros.qmd` contains only raw LaTeX macro definitions (`\def`, `\providecommand`, etc.), Quarto passes them through as raw LaTeX for PDF output (where they are processed inline in the document body as LaTeX commands) and as math for HTML/RevealJS output (where they are processed by MathJax).

Note: This demo uses `macros.qmd` directly (same directory). When using this repository as a submodule, use `macros/macros.qmd` instead.

Note (MathJax limitations): [Quarto's MathJax integration](#) supports `\def`, `\newcommand`, `\renewcommand`, `\newenvironment`, `\renewenvironment`, and `\let` — but **not** `\providecommand`. Since `macros.qmd` uses `\providecommand` for many definitions, those macros are silently ignored for HTML/RevealJS output. The example table below uses only `\def`-based macros, which render correctly in all formats.

Proof that `\providecommand` is ignored in HTML/RevealJS

The macro `\floor` in `macros.qmd` is defined with `\providecommand`. Compare this with a `\def` macro:

Macro type	Code	Output
<code>\providecommand macro (\floor)</code>	<code>\$\$\floor{1.2}\$\$</code>	$[1.2]$
<code>\def macro (\a)</code>	<code>\$\$\a\$\$</code>	α

In HTML/RevealJS, `\floor` remains unresolved while `\a` renders correctly, showing that `\providecommand` definitions are ignored by MathJax.

Note (RevealJS blank slide): Content placed before the first `##` heading in RevealJS output becomes a blank title slide. Since the include shortcode is at the top of this file, RevealJS may show a blank slide before the first section. This is a [known limitation](#) of the shortcode approach for RevealJS. For RevealJS without a blank slide, use the [include-in-header strategy](#) instead.

Example math

The table below uses macros defined with `\def` in `macros.qmd`. These `\def`-based macros work across all output formats — note that `\providecommand`-based macros from `macros.qmd` are silently ignored for HTML/RevealJS as noted above:

Description	Code	Output
Greek: alpha	<code>\$\$\a\$</code>	α
Greek: beta	<code>\$\$\b\$</code>	β
Greek: gamma	<code>\$\$\g\$</code>	γ
Greek: lambda	<code>\$\$\l\$</code>	λ
Greek: sigma	<code>\$\$\s\$</code>	σ
Log-likelihood	<code>\$\$\llik\$</code>	ℓ
Independent	<code>\$\$X \ind Y\$</code>	$X \perp\!\!\!\perp Y$
IID distributed	<code>\$\$X_i \siid \Normal(\m, \ss)\$</code>	$X_i \sim_{\text{iid}} \text{N}(\mu, \sigma^2)$
Vector: mu	<code>\$\$\vm\$</code>	$\tilde{\mu}$
Hat beta	<code>\$\$\hb\$</code>	$\hat{\beta}$